

SSH Banner with Ansible

Ad-Hoc Commands Lab

Safe tareeqay se configure, validate, reload, verify aur rollback karein

TARGET

three_tier_app

CONTROL NODE

ansible-server

MANAGED NODES

node1 node2 node3

Learning path

SSH configuration change ka safe sequence

- 01** Target aur modules ko samjhein
- 02** Banner file create karein
- 03** Regex se sshd_config manage karein
- 04** Validate, reload aur test karein
- 05** Idempotency check aur rollback karein

Lab environment

Rocky Linux control node aur teen managed nodes

CONTROL NODE

ansible-server

192.168.1.233

ansibleadmin

```
/home/ansibleadmin/  
automation
```

MANAGED NODES

node1

192.168.1.154

Rocky Linux

node2

192.168.1.185

Rocky Linux

node3

192.168.1.190

Rocky Linux

SSH change se pehle safety rules

Configuration test karte waqt apna access mehfooz rakhein

1 Current SSH session open rakhein

2 Testing ke liye second terminal use karein

3 Pehle --limit node1 use karein

4 Edit se pehle backup banayein

5 Reload se pehle validate karein

6 Restart ke bajaye reload ko tarjeeh dein

Tip: `--limit node1`

Is lab mein use hone wale modules

Har module workflow ka ek khaas hissa manage karta hai

ping

Ansible connectivity test karta hai

copy

Banner file create karta hai

command

Shell ke baghair safe command
chalata hai

stat

File metadata check karta hai

lineinfile

Banner directive manage karta hai

shell

Pipe ke sath effective config check
karta hai

service

sshd reload karta hai

file

Cleanup mein banner file remove
karta hai

Project configuration aur connectivity

/home/ansibleadmin/automation se commands chalayein

```
[defaults]
inventory = ./inventory/nodes
remote_user = ansibleadmin
private_key_file = /home/ansibleadmin/.ssh/ansible-
key
host_key_checking = False
```

```
[privilege_escalation]
become_method = sudo
become_user = root
become_ask_pass = False
```

```
cd /home/ansibleadmin/automation
ansible three_tier_app -m ping
ansible three_tier_app --list-hosts
```

SSH banner create karein

Template variable har host ko unique message deta hai

```
ansible three_tier_app -b -m copy -a \
'content="*****\nWelcome to {{ inventory_hostname }} for Ansible
practice\n*****\n" dest=/etc/ssh/banner.txt owner=root group=root mode=0644'
```

HAR NODE PAR RESULT

Welcome to node1 for Ansible practice

Banner file inspect karein

Content ke liye command aur metadata ke liye stat

CONTENT

```
ansible three_tier_app -b -m command -a \  
"cat /etc/ssh/banner.txt"
```

changed: false

METADATA

```
ansible three_tier_app -b -m stat -a \  
"path=/etc/ssh/banner.txt"
```

owner=root mode=0644

sshd_config manage karein

lineinfile commented ya active Banner directive replace karta hai

```
ansible three_tier_app -b --limit node1 -m lineinfile -a \
'path=/etc/ssh/sshd_config \
regexp="^\s*#\s*Banner\s+.*$" \
line="Banner /etc/ssh/banner.txt" state=present backup=yes \
validate="/usr/sbin/sshd -t -f %s"'
```

DESIRED STATE

Banner /etc/ssh/banner.txt

node1 successful ho to command --limit ke baghair repeat karein.

Regex ki anatomy

Pattern ko left se right parhein

```
^\s*#?\s*Banner\s+.*$
```

^

Line ka start

\s*

Zero ya zyada spaces

#?

Optional # comment

\s*

Dobara optional spaces

Banner

Exact directive name

\s+

Ek ya zyada spaces

.*

Baqi tamam value

\$

Line ka end

Regex kya match karta hai

— Anchors partial directive names ko match hone se rokte hain

MATCH HOTA HAI

```
#Banner none
# Banner /etc/issue.net
Banner none
    Banner /etc/ssh/banner.txt
```

`^ ... $`

MATCH NAHI HOTA

```
BannerFile /tmp/banner.txt
MyBanner none
AllowBanner yes
```

Directive name exact nahi hai

Multiple matching lines

lineinfile aam tor par aakhri match change karta hai

Agar sshd_config mein kai Banner lines hon, lineinfile aakhri matching line ko change karta hai. Har match ko change ya remove karna ho to replace module use karein.

Complete SSH configuration validate karein

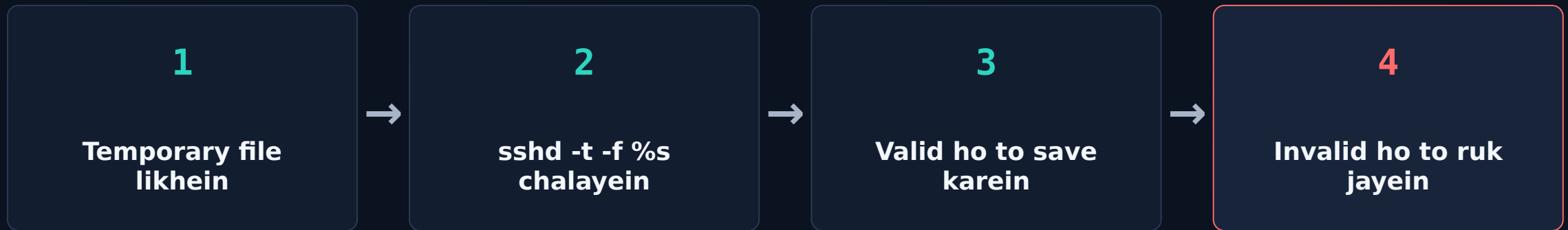
```
ansible three_tier_app -b -m command -a "/usr/sbin/sshd -t"
```

```
ansible three_tier_app -b -m shell -a \  
"/usr/sbin/sshd -T | grep '^banner '"
```

Apply karne se pehle validate

Ansible pehle temporary configuration file test karta hai

SAFE FLOW



```
validate="/usr/sbin/sshd -t -f %s"
```

sshd reload aur verify karein

Valid configuration apply karein, existing sessions ko drop kiye baghair

```
ansible three_tier_app -b -m service -a \  
"name=sshd state=reloaded"
```

```
ansible three_tier_app -m command -a \  
"systemctl is-active sshd"
```

```
ansible three_tier_app -m shell -a \  
"pgrep -af sshd"
```

SSH client se test karein

Banner authentication se pehle nazar aata hai

```
ssh -i ~/.ssh/ansible-key ansibleadmin@node1  
ssh -i ~/.ssh/ansible-key ansibleadmin@node2  
ssh -i ~/.ssh/ansible-key ansibleadmin@node3
```

EXPECTED EXPERIENCE

Banner → authentication prompt → successful login ke baad shell

Idempotency

Doosri run par dobara change nahi hona chahiye

PEHLI RUN

changed: true

DOOSRI RUN

changed: false

copy aur lineinfile commands dobara chalayein. Stable configuration par change nahi hona chahiye.

Rollback aur cleanup

SSH ko original state mein wapas laayein aur lab artifacts remove karein

```
ansible three_tier_app -b -m lineinfile -a \
'path=/etc/ssh/sshd_config regexp="^\s*#?\s*Banner\s+.*$" \
line="#Banner none" state=present backup=yes \
validate="/usr/sbin/sshd -t -f %s"'
```

```
ansible three_tier_app -b -m service -a "name=sshd state=reloaded"
ansible three_tier_app -b -m file -a "path=/etc/ssh/banner.txt state=absent"
```

Rollback ke baad SSH dobara test karein.

Review checklist

Mukammal ad-hoc configuration workflow



Connectivity confirm hui



Banner file create hui



Regex se directive manage hua



Configuration validate hui



sshd reload aur check hua



SSH login test hua



Second run idempotent hai



Rollback complete hua